

# User Guide for Sinaps Counterbalance sys

## Rotation System and tension control

The commutation should rotate the same direction as target's rotation(Rat) whenever the degree difference accumulates to 90 degrees.

### 1. Connection check

There are 3 connection on PCB board

- USB port between PC and Arduino nano
- Motor power
- Motor and driver(A4988)

Ensure they are all connected.

### 2. Software check

1. Turn on cameras and Motive
2. Create rigidbody and set streaming ID
3. Enable streaming
4. Open motivetracking.py  
(under Desktop/NatnetSDK/Samples/pythonclient/)
5. Run the code, if the connection works, it will show port detected.

Should show:

```
busy=yes, Δyaw={delta_yaw:+.2f}°, accumulation={accum_angle:+.2f}°"
```

Common Question:

1. **All plugs are connected and you hear that motor is moving but not rotation** correctly- check the connection of the motor and driver, the 4 pins might be inverse.

## Counterbalance system

This closed loop system is independent of the rotation system.

### 1. Check hardware connection

1. battery for motor (DO NOT attach inversely GND and 9V)  
the LED on DRV8833 should turn on

### 2. plug on the headstage to animal

### 3. Pull the thread to make it not lean on the load cell

### 4.. plug on the power of the Arduino nano

### 5. release the thread

To know,

### 1. highly recommend to go through the process without animal first

### 2. **Once the two power connections are plugged, the sys will start to work in closed-loop, please do the safety check on connection to the animal first**

### 3. the parameters need to be adjust depend on the weight of the cable and where it attach

Parameter for counterbalance system:

In file /code/counterbalce/auto.ino, you can play with the parameter for customized env.

### **Sensor parameters:**

Line 71-72

Lower limit- Increase if the wire is too loose but sys didn't retract

-Decrease if the wire is already tight but still retracting

Upper limit-Increase if the sys is too sensitive to the jittering

-Decrease if feel too strong resistance when pulling

### **Motor parameters:**

Retracting :

Line 161- Deadzone :The minimum PWM that can drive the motor

Line 37-40-speed of the retracting

Releasing:

43- Base\_payout: minimum speed when the tension cross the boarder(upper\_threshold)

**44- Maxpayout:** this is the key that should be control. If the sys release to fast in a very short period, the wire will be over released and too long that will cause entangle.

48.Burst\_payout: when the tension increase sharply, the speed of the payout, is constrained by 44.

Recommended numbers for sinaps.

Line/numbers

71/72:14.8/32.5 (can adjust in the step of 0.5g)

161:70

43:70

44:110 or 120